

Architectural Particular Specification
ROLLER SHUTTERS
SECTION 10.3

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SECTION 10.3

ROLLER SHUTTERS

PART 1 GENERAL

1.01 SUMMARY OF WORK

- A. To design supply, install and test all roller shutters as per the shutter schedule on the drawings. This shall include all motor equipment, operator, ironmongery, assembly jointing, gears, boxes, hoods, guides, fixing lugs, bracket, cutting and pinning lugs, access panel/doors (FRP/non-FRP) for shutter maintenance and other necessary fixings and parts.
- B. The dimensions of the shutters shown on the schedule shall be used as a reference. The contractor shall verify on site prior to preparation of shop drawings.
- C. The design of the components, assembly, and operation system of the roller shutters, including all motor equipment, gears, boxes, hoods, guides, and other parts shall be the responsibility of the Contractor.
- D. The Contractor shall, jointly with the manufacture and/or specialist supplier/installer, provide all necessary details, drawings and information for submissions to satisfy relevant requirements by other relevant government departments.
- E. The Contractor shall co-ordinate with the manufacturer and/or specialist supplier/installer for all essential services required for the shutters.
- F. The Contractor is also responsible for providing and installing all requisite builder's works in association with the proper installation and operation of the roller / fire shutters as well as the fire / smoke curtain. This shall include, without limitation to:-
 - 1. The construction of brick / blockwork jamb for accommodating the vertical channel guides on both sides of the shutter. Such brick / blockwork shutter jamb shall be in matching finish with the adjoining wall surfaces;
 - 2. The construction of hanger wall / FRP infill panel above the fire shutter.

1.02 REFERENCE

- A. See ASD GS 2012 clause 17.43.

1.03 QUALITY ASSURANCE

(NOT USED)

1.04 PERFORMANCE REQUIREMENTS

- A. Refer ASD GS 2012 clauses 17.41 and the following.
- B. Unless otherwise shown in Drawings, all shutters shall be as follows:-
 - 1. Electrically operated with fail safe and manual-override device(s);
 - 2. Vertical channel guides, hoods, key switch control boxes and casings shall be stainless steel or of materials specified in the drawings;
 - 3. Exposed / visible parts shall be Hairline stainless steel finish;
 - 4. Of minimum 1.2mm thickness for slat;
 - 5. Supplied and fixed by experienced specialist installer(s).

C. Fire Resistant Shutter

1. See ASD GS 2012 clause 17.42;
 2. Those fire shutters together with their accessories forming a complete system should have a FRP as specified in the drawings. The Contractor shall, jointly with the manufacturer and/or specialist supplier/installer, prepare relevant test reports and certificates from recognized laboratories under the HOKLAS scheme showing the conformity of the system with BS 476:Part 20 to 24:, the Code of Practice on Fire Resisting Construction 1996 issued by the Buildings Department, and the latest requirements of Fire Services Department;
 3. Design Criteria to be BS 476 : Part 22 : 1987.
- D. All the fire shutters shall be connected to the central fire alarm system and essential power supply.
- E. Hoods and casings shall be of materials specified in the drawings and of adequate gauge fully supported and braced to prevent any sagging or distortion. Access doors or plates shall be provided at the appropriate places to enable routine inspection and servicing to be carried out.
- F. The entire shutter system(s) including materials, structure, installation, workmanship and performance to be in proper and normal function in accordance with the contract requirements for a period of not less than FIVE years from the date of substantial completion as certified by the Architect.
- G. The key switch box & break glass unit should be watertight if it is located near wet or outdoor area. Works shall conform to all relevant statutory ordinances, regulations, orders, codes of practice in force at the time of signing of the Contract.
- H. The Contractor shall allow for site tolerances, discrepancies and changes in the building layout, where the principles and intent of the signs are not affected.

1.05 ENVIRONMENTAL REQUIREMENTS

(NOT USED)

1.06 ENVIRONMENTAL, CLIMATIC & PSYCHOMETRIC CONDITIONS

(NOT USED)

1.07 SUBMITTALS

- A. See ASD GS 2012 clauses 17.14 & 17.44.
- B. Shop Drawings

Manufacturer's catalogues, test certificate, and installation method, material samples and shop drawings for all relevant materials and equipments shall be submitted to the Architect for review or confirmation of no adverse comment. The extent of information or samples required shall be adequate and sufficient to demonstrate that the proposed system and materials are in compliance with the contract requirements. However, the Architect can request the contractor to submit any additional or supplementary information to substantiate the performance of the proposed system or material.

C. Samples

Materials and finishes of slat, guide rail, bottom bar of vertical shutter and receiving guide, leading edge and housing of horizontal slide shutters shall be submitted to Architect for approval.

- D. The Contractor shall verify all dimensions as specified in the shutter schedule on site and prepare the shop drawing as per the measured dimensions.
- E. Specimens of paint colour and primer shall be submitted to the Architect for selection and approval before proceeding with the work.
- F. Upon completion of work, contractor shall supply the approved OPERATION AND MAINTENANCE MANUAL (instead of “two instruction manual” stated in ASD GS 2012 clause 17.44) including all circuit diagrams, schematic diagrams, approved equipment shop drawing, schedule of component parts, priced list of recommended spares, and other technical information provided by the manufacturer. A draft of the manual shall be submitted in advance to the Architect for review and approval before final submission.

1.08 PRE-INSTALLATION CONFERENCE

(NOT USED)

1.09 DELIVERY STORAGE AND HANDLING

Deliver materials in original packaging supplied by manufacturer with intact labels. Store materials away from harmful environmental conditions and construction.

1.10 MOCK-UPS

This part shall be read in conjunction with Clause 8.03 of Specification Preliminaries for the requirements for prototypes and trial areas.

1.11 TESTING

(NOT USED)

1.12 WARRANTY

This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.

1.13 SPARE PARTS

This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.

PART 2 PRODUCT

2.01 APPROVED MANUFACTURER

Subject to compliance with the Project Contract Documents, provide roller shutter system and all required accessories from one or more of the following approved manufacturers/suppliers, or equivalent approved by the Architect:

1. Sherex Engineering Limited
Office Address: Flat A & C, 10/F., Lucky Horse Ind. Bldg., 64 Tong Mi Road,
Mongkok, Kowloon, Hong Kong
Tel : (852) 2688 6213 Fax : (852) 2787 9633
Email: sales@sherex.com.hk

2.02 MATERIALS

A. Vertical Insulated Fire Shutter

1. General
 - a. Insulated fabric fire shutters shall be housed inside insulated shutter boxes and shall be operated along guide rails;
 - b. For vertical rolling insulated fabric fire shutters, the shutters shall be rolled on barrels inside insulated shutter boxes.
 - c. Insulated shutter box shall be used for housing insulated fabric fire shutters, barrels and motors;
 - d. For vertical rolling insulated fabric fire shutters, guide rails shall be provided on both sides of the fire shutter and shall be used to guide the movement and operations of shutter;
 - e. All insulated fabric fire shutters shall be driven by motors associated with motor control units, master interface units and 2 manual operation units.
 - f. The installation shall fully comply with the "Code of Practice of installation of Electric Operated Sliding Gates, Sliding Glass Doors and Roller Shutters" published by Hong Kong Electrical and Mechanical Service Department, Fire Services Department's requirements, Buildings Department's requirement and any requirements issued by relevant authorities.
2. Fire Rating
120 mins Integrity and 30 mins Insulation
3. Insulated vertical fabric fire shutter
 - a. The insulation performance of the insulated fabric fire shutters shall be tested and certificated to the standard in accordance with BS EN 1634-1; i.e. the temperature rise at any one point on the unexposed surface of the shutter around all shutter box outlets, guide rail outlets, receiving channel inlets, bottom bar floor gaps, bottom driving chain floor gaps and overlapping shutter joints shall not be more than 180OC in the fire test;
 - b. 30 minutes, 60 minutes and 120 minutes insulated fabric fire shutter shall be formed by single layer, while 240 minutes insulated fabric fire shutter shall be formed by double layers.
 - c. The shutter hood shall be manufactured from galvanized mild steel. The enclosure shall be rated same with the shutter fabric with corresponding integrity and insulation. Removable cover plate shall be incorporated to allow access to the shutter.
 - d. For insulated fabric fire shutters of 240 minutes insulation fire resisting rating, the two fire shutters shall not be more than 100 mm apart from each other;

- e. A motor control panel which housed in a steel enclosure shall be mounted near the fire shutter within 3 meters.
 - f. The shutter shall be manufactured from composite fire resisting materials incorporating stainless steel wire mesh covering with glass fiber cloth and intumescent composite material with nominal 18mm thickness for 30 minutes insulated fabric, nominal 27mm thickness for 60 minutes insulated fabric, nominal 38mm thickness for 120 minutes and nominal 11.8mm for each layer of 240 minutes insulated curtain fabrics.
 - g. The shutter curtain fabric shall have a nominal weight of 7kg/square meter (30 minutes insulation), 8kg/square meter (60 minutes insulation), 10kg/square meter (120 minutes insulation) and 5.2kg/square meter each layer (240 minutes insulation), and shall be tested to withstand temperatures up to the required fire resistance rating for both insulation and integrity.
 - h. The insulated vertical rolling fabric shutters shall be fixed onto a barrel made of seam welded galvanized mild steel tube of calculated size and thickness to suit the width of the opening and to support the weight of the shutter and the accessories.
 - i. All insulated vertical rolling fabric shutter doors shall be provided with fabric retention tags and metal rings on both sides of the shutter door at a space of not more than 250 mm center to center. The fabric retention tags and metal rings shall be able to secure the shutter door onto and running along the guide rails
4. Overlapping insulated fabric fire shutters
- a. For an vertical insulated fabric fire shutter more than 8 meters in width, it shall be constructed of overlapping design;
 - b. The overlapping portion of the shutter shall be not less than 600 mm in width;
 - c. Each section of the shutter shall be fixed onto individual barrels which shall be arranged in stagger positions either vertically or horizontally and driven by individual motor;
 - d. All sections of the overlapping shutter shall be linked up by a single bottom bar and shall be operated in a synchronized movement as a one system.
5. Shutter box construction for overlapping insulated fabric fire shutters
- a. For overlapping insulated fabric fire shutters, the shutter box shall be constructed of composite insulated fire rated boards;
 - b. The composite insulated fire rated boards shall be composed of galvanized mild steel as outer shell of not less than 1.5 mm in thickness and shall be sandwiched with core insulated material with total thickness of not less than 23 mm;
 - c. The core insulated materials shall be not less than 20 mm in thickness and 380 kg/m³ in density;
 - d. The end plates and intermediate end plates of the shutter box shall be designed and used to support the barrel and shall be minimum 5 mm thick galvanized mild steel as required to safely transmit the weight of shutter and accessories.
 - e. Each shutter box shall be capable of housing multiple shutter which shall be arranged in staggered positions.
6. Operation
- a. The fire shutter shall deploy upon a signal from the fire alarm system in an emergency.

- b. Under normal operating conditions the shutter would be held in the retracted position by built in braking system.
- c. Upon activation of the fire alarm the control panel will operate the shutter to descend in a controlled manner. A dynamic braking system housed in the motor control circuit shall control the speed of the descent of the shutter. The descent shall be electronically synchronized on overlapping shutters with a bottom bar.
- d. To retract the shutter, the control panel shall activate to the motor control circuits and motor will drive the shutter to the upper position. As the bottom bar or stopping bar reach the preset upper limit, the motor will stop automatically and hold the bottom bar in the retracted position.
- e. Limit switches will be used to control the upper and lower position of the shutter.
- f. For fire shutter, manual control units incorporating ascending, stop and descending buttons shall be provided. The units shall be housed inside a stainless-steel metal box with a breakable glass front.

7. Components

a. Barrel and Motor

- 1) The specimen consisted of an electric motor and a barrel shaft. The barrel is fixed at each end to a bearing, which are fixed onto an end plate. Each end plate is fixed into the concrete support frame by anchor bolts.
- 2) The barrel and motor are protected by a barrel protection made of steel angles anchored to the concrete support frame.
- 3) 2 layers of Durasteel plates or other equivalent material shall be fixed along the steel angles by screw to provide FRR enclosure as required.

b. Shutter Complex

- 1) The rolling shutter shall be a multi-layer complex by a metal lath layer and sandwiched on both side by a multilayer fabric cloth layer.
- 2) The metal lath made of folded metal sheets with mineral fiber infill. Both ends of the metal lath are constructed with a return to interlink with other laths.
- 3) Both sides of the metal lath layer shall be lined with a sheet of multilayer fabric cloth. The silvery surfaces of the fabric are exposed whereas the grey surfaces are facing the metal slats. The fabrics are clamped by metal stiffeners and nut-and-bolts through.

c. Guide Rails

- 1) The guide rails shall be made of stainless steel or galvanized mild steel (GMS) profile.
- 2) Fire resistance board shall be concealed between the outer and inner sheath and intumescent material shall be installed along the inner face of guide rails to suit the F.R.R. as required.
- 3) The guide rails are fixed by welding to steel angle brackets which are previously fixed into the concrete support frame by anchor bolts.

d. End Plates

- 1) Roller barrel to be supported between two end plates of minimum 3mm thick GMS as required to safely transmit the shutter weight and according to the shutter width to determine the thickness of the end plates.
- 2) Fixings to structure shall be GMS and of sufficient size and number to support the shutter.

e. Bottom Bars

- 1) The bottom bars shall be installed at the bottom of shutter curtains. The bottom bar of exposed fabric shutter shall be made of GMS / SS angles, which are fixed to the curtains both sides by bolt-and-nut through.
- 2) The bolt-and-nuts fixings of bottom bars on both curtains are distanced at approximate 600mm centre to centre.

f. Motor

- 1) Shutter motor shall be mounted on the upper level of the shutter coil box for shutter operation.
- 2) Speed range for electrically operated shutter, as tested by manufacturer's own method. The descending speed should be complied with EMSD's requirement within 200mm per second and separated two compartments by 15sec -60 sec according to FSD's requirement.
- 3) Activation or descent of the FIRE RESISTANCE SHUTTER system shall raise a common fire alarm on the Automatic Fire Alarm panel.
- 4) The power supply to be 1 phase, 220V 32A, essential power which provided by MEP contractor and located within 2m of the shutter motor.
- 5) Provide crank-hoist operator consisting of crank and crank gearbox, steel crank drive shaft, and gear-reduction unit.
- 6) Fabricate gearbox to completely enclose operating mechanism and be oil tight.
- 7) Provide manufacturer's standard crank-locking device.

g. Push button control cabinet

- 1) The shutter control push button contained "UP", "STOP" & "DOWN" button inside the cabinet.
- 2) The cover plate to be stainless steel material as dimension in 150mm H x 90mm L completed with break glass and key lock. It allowed to press the button unless emergency purpose.

8. Approved Manufacturer and Supplier

Brand Name: "Forbes" or approved equivalent

Supplier: Sherex Engineering Limited

Office Address: Flat A & C, 10/F., Lucky Horse Ind. Bldg., 64 Tong Mi Road, Mongkok. Kowloon, Hong Kong

Tel: (852) 2688 6213 Fax: (852) 2787 9633

Email: sales@sherex.com.hk

B. Motorized Perforated Steel Roller Shutter for Ring Main Unit (RMU) Bays

1. Material

- a. Made from strong interlocking galvanised steel slats, perforated throughout with small airholes
- b. Interlocking curls solid and strong, making for stronger curtain
- c. Slat width: 50mm / 75mm
- d. Allow at least 50% overall airflow, but can combine with solid slats for control of airflow and vision
- e. Shutter curtain shall be finished with protective coating powdercoat (applied during manufacture)

2. Accessories
 - a. Head panels
 - b. Pedestrian egress gates
 - c. Weatherstrips
 - d. Automatic operation through logic control box
 - e. Key operated isolating switch to enhance security
 - f. Outside key switch for external operation of door
3. Operation
 - a. By Motor – using standard or high cycle rated motor units according to door usage. (see Technical Information for details)
4. Technical information
 - a. Roller:

Seamless drawn steel tube housing counterbalanced springs
 - b. Roller Axle:

Solid mild steel
 - c. Slat:
 - 1) 50mm x .8mm profile with nylon end clips; or
 - 2) 75mm x .9mm profile with nylon end clips
 - d. Guide:
 - 1) 64mm x 35mm x 2.5mm galvanised
 - 2) 75mm x 35mm x 2.5mm galvanised
 - e. Bottom Rail:
 - 1) 50mm x 50mm x 6mm galvanised steel angle
 - 2) 50mm x 50mm x 6mm aluminium angle
 - 3) Brackets: 8mm mild steel plate
5. Motor: Standard 3ph 415v 1HP (.75kw) motor unit fitted with emergency hand chain and 240V reversing direct on line push button switch
6. Finish
 - a. Powdercoat (pre-coated); or
 - b. Polyurethane (painted on site)
7. Motorised: Locked automatically by gearbox

C. Security Vehicle Shutter

Requirements refer to Technical Specification for security system design.

D. Typhoon Resistant Shutter

1. The contractor shall demonstrate that the shutter can stand the possible wind pressure in Hong Kong. The ends of all slats shall be fitted with stainless steel end stops riveted on to prevent lateral movement and to protect the shutter against wear.

E. Vision Shutter

1. Shall have white powder-coated aluminum vertical drop panel/slat with aluminum hinge, built-in curtain hold-fast device, and 3mm thick UV protected multiple polycarbonate panels system.

2. Flammability of polycarbonate panel to UL94-V0.
3. Panel to resist artificial daylight and sunlight up to at least 3 years without any discoloration or clarity changes, and shall be able to ensure sunlight, high intensity spotlight at 60deg C without change in color/texture/appearance.

PART 3 EXECUTION

3.01 EXAMINATION / INSPECTION

(NOT USED)

3.02 PREPARATION

- A. Preparation for opening and installation of fire door to be in strict compliance with COP of Fire Safety in Building 2011.

3.03 INSTALLATION / APPLICATION

- A. Install rolling shutters in accordance with manufacturer's instructions.
- B. Ensure that units are installed plumb and true, free of warp or twist, and within tolerances specified by manufacturer for smooth operation.

3.04 TOLERANCES

(NOT USED)

3.05 CLEANING

- A. Upon completion of the work, all equipment shall be properly cleaned, greased and protected from weather. Before handing over, the contractor shall instruct the representative of the Employer in operation of the equipment until he is familiar with the same.
- B. The Contractor shall be responsible for making good any defect which may appear after the rolling shutters shall have been taken over, and which arises either from defective materials, workmanship or design.
- C. Provide two complete sets of necessary special tools required for each shutter shall be provided at appropriate location to facilitate manual operation and future maintenance.

3.06 PROTECTION

(NOT USED)

END OF SECTION